

Lab 02

LINEAR REGRESSION (OLS)

CODE

```
x = [10,8,7,4,6]
y = [50,40,42,12,18]

x_mean = sum(x) / len(x)
y_mean = sum(y) / len(y)

num = 0
den = 0

for i in range(len(x)):
    num += (x[i] - x_mean) * (y[i] - y_mean)
    den += (x[i] - x_mean) ** 2

m = num / den
b = y_mean - m * x_mean

print("Slope =", m)
print("Intercept =", b)

# Prediction
x_test = 4
y_pred = m * x_test + b

print("Prediction =", y_pred)
```

OUTPUT

```
● PS E:\class> python -u "e:\class\linear_withoutgradient.py"
Slope = 6.8
Intercept = -15.200000000000003
Prediction = 11.999999999999996
○ PS E:\class>
```